this fuel planet. It will require carbon dioxide and water locally. It is beyond this stage that things get increasingly improbable. SpaceX is planning a manned mission to Mars with a crew of 12 to make sure the propellant plant is running properly. The earliest pioneering missions by NASA is also beyond the 2030 horizon. We are not saying the SpaceX base will not happen, but expect some delays.

ARTIFICIAL GENERAL INTELLIGENCE

There is no question about it, Artificial Intelligence will own the next decade. According to Sundar Pichai, the next decade belongs to AI. In the mid 80s, desktop computers changed everything, in the mid 90s, it was the internet, and in the mid 2000s, we saw the advent of smartphones. It is the mid 2010s now, and according to Google, this will totally be AI. IBM, Microsoft, Baidu and other tech majors are banking on AI as well. Artificial general intelligence, is however the point at which an AI can do the entire range of tasks that a human can. The problem here is not a specific task, which AI is already better at in many areas. The challenge here is a single AI being able to understand, adapt, and provide a superior response to a human, every single time, when faced with a wide variety of tasks. This kind of AI is called "strong AI" or "full AI". The incipient field of neuromorphic computing will have to mature first, which are computers built in the same way that human brains are. We still do not have the raw computing power that artificial general intelligences will require to do that.

CROSS SPECIES COMMUNICATION

Species already communicate between themselves. This is about technological interventions to enable cross species communication. While there are already technological solutions to better understand what pets want, or make more accurate guesses, we are a far way away from true cross species communication. This is because, our brains are all wired differently, and our understanding of the world is considerably different. For example, dogs depend far more on the sense of smell than humans do. It would be hilarious, and useful to directly understand what your pet is thinking when it is introduced to novel environments and people. To realise cross species communication, we need to get neural implants that understand all the signals in the brain. Then we need machine learning algorithms to translate the intentions and desires of one species, and make them comprehensible to another species. This is just to communicate with humans. Getting your pet parrot to communicate with your pet cat in a meaningful manner, is nowhere close to realisation over the course of the next decade. A cross species communication system is a far cry.

3D PRINTING

3D printing, or additive manufacturing has been around for a long time. The first systems were actually created in the 1980s. Even after almost 40 years, it is nowhere close to becoming a consumer grade technology. This is because the end products are not very useful or durable. It also takes a long time to print, and it is expensive as well. Additive manufacturing will continue to grow, but is more likely to be used by services that cater to very specific markets. This could be a 3D printing studio for sundry tasks for artists, architects, and cosplayers. It could also be garages that produce custom spare parts for cars, and introduced in OEM assembly lines. Consumer level 3D printing has not happened for the last forty years, and we don't see how it could possibly happen in the next ten.

MULTIPURPOSE HOUSEHOLD ROBOTS

The challenges with building multipurpose household robots is everything facing artificial general intelligence, plus the robotic implementation. For example, it is very simple to make a robot drive a car, open a door, walk a gangway, or turn a valve. In fact, this can be done just by having robotic cars, doors, and valves. The problem is a single robot that can perform all of these tasks. That would require a roughly human form factor, which robots are very bad at. These robots would be required to use the same appliances and devices that are designed to be used by humans. An android robot cannot even open a door or smash an axe through one, without toppling over. Boston Dynamics is one of the leading companies in this field. The thing is, we might not even need a general purpose household automaton, such as Rosie the Robot in Jetsons. Household automation, IoT, smart speakers can together accomplish separately, what a household robot could possibly do. Forget not happening over the next decade, this one we are chalking down to not even necessary.

DRONE DELIVERY

Of all the things in the list, this is actually the most likely to come true. Both Amazon and UPS are testing out products that use drones to deliver parcels. The problem is scaling up the service, to build a worldwide network. There are many different variables that autonomous drones will have to handle, and they might not be ready to tackle all these situations within the next ten years. There are limitations on the weight and the distance for which a drone can make a delivery. Landing and taking off is a tricky process, and companies are exploring options such as using parachutes and specially marked drop zones. To allow drone delivery at scale, they will need to be proven safe enough, and pass a number of regulatory hurdles. The on-board cameras and sensors, for example, will have to prove that they do not invade the privacy of the people. Then there is the question of security, when dispatching high value items such as mobile phones. While drone deliveries will certainly implemented in certain pockets, we don't see a global implementation happening seamlessly within the next decade.